

DYNAMICS GUARD Technical Manual

Envelope, transient, pressure, and containment control

Control entry, impact, sustain, and pressure without hiding the source.

Included products: ANUBIS, OTIS, DEBO, LOOKOUT

Routing authority: Use on individual tracks or buses according to Smart Position. DEBO may be used for parallel impact.

Canon baseline

Normalized control 1 equals 0.809016994 (PHI/2). Standard knob travel is 7 o'clock to 5 o'clock. Factory programs must be clickless and must honor protected I/O, Mix, Power, and ceiling controls. Smart Position remains the placement authority.

ANUBIS

Dynamics gate and guard with threshold, ratio, attack, hold, release, range, a named stereo host Sidechain bus, independent Sidechain Key and Sidechain Duck operation, protected I/O, and parallel mix.

Declared product: ANUBIS | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Insert Safe | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call not detected; integration review required before release claims
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
input	Input	Float
mix	Mix	Float
output	Output	Float
power	Power	Bool
threshold	Threshold	Float
ratio	Ratio	Float
attack	Attack	Float
hold	Hold	Float
release	Release	Float
range	Range	Float
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice
extkey	Sidechain Key	Bool
scduck	Sidechain Duck	Bool
gainmatch	Gain Match	Bool

Artwork authority

Assets/Images/Anubis.png

Meter and visualizer semantics

Display	What it measures	How to read it
IN	Main-input RMS energy	Shows the program feeding the gate.
GATE	Current gate passage gain	OPEN is full passage; lower values mean attenuation.
KEY	External sidechain RMS energy	Moves only when a host sidechain is routed.
DUCK	Sidechain-driven gain reduction	More fill means deeper ducking of the main signal.
OUT	Post-gate RMS energy	Compare with IN to judge actual attenuation.
Transfer graph	Threshold, envelope, and Range floor	Shows where the signal opens and how far closure may fall.
Correlation waves	Left/right phase relationship	Gold is coherent, purple is spread, red warns of anti-phase risk.

ANUBIS exposes a named optional stereo host Sidechain bus. Sidechain Key routes that bus to the detector/envelope while the main input remains the only audible program path; Sidechain Duck independently applies the detected envelope as ducking.

OTIS

3-6-9 macro dynamics processor: Micro Peak controls short events, Meso Density shapes body, and Macro Level governs the long envelope.

Declared product: OTIS | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Insert Safe | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
trumpet_l	Micro Peak	Float
trumpet_c	Meso Density	Float
trumpet_r	Macro Level	Float
makeup_gain	Makeup Gain	Float
gainmatch	Gain Match	Bool

Artwork authority

source/Otis.png

Meter and visualizer semantics

Display	What it measures	How to read it
INPUT	Input peak in dB	Confirms level feeding all three time scales.
RESPONSE	Strongest active 3/6/9 reduction	Shows the greatest current reduction across Micro, Meso, or Macro.
OUTPUT	Output peak in dB	Use with INPUT and Gain Match for fair comparison.
3 / Micro activity	Fast transient reduction	Responds to short peaks and front edges.
6 / Meso activity	Body and phrase-density reduction	Responds to medium-duration energy.
9 / Macro activity	Long-envelope reduction	Responds to sustained level movement.

Factory programs

OTIS PRESET

DEBO

Pressure-activated dynamics processor with nine updated canonical programs using the restored iCloud names. The approved powered-on default is unchanged; the named bank uses a warmer depth/pressure progression without runtime preset clamping, topping out at 2.2 dB Maximize.

Declared product: DeBo | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Insert Safe | **Channel mode:** not detected | **Dual-mono link:** not detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call not detected; ceiling review required
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
POWER	Power	Bool
FLOWER	Flower of Life	Bool
GAINMATCH	Gain Match	Bool
INPUT	Input	Factory
DRIVE	Drive	Factory
MAXIMIZE	Maximize	Factory
DEPTH	Depth	Factory
CLARITY	Clarity	Factory
OUTPUT	Output	Factory

Artwork authority

Assets/Images/Debo_state1_emerge.png

Meter and visualizer semantics

Display	What it measures	How to read it
RMS	Sustained signal body	Shows average energy driving the pressure model.
GUARD PRESSURE	Combined density/containment demand	Higher values mean DEBO is working harder.
PEAK	Fast peak relative to ceiling	Approaching the top warns of limited remaining space.
GOVERNOR	Gain held back by protection	More activity means stronger safety containment.
Pressure window / face	RMS-driven pressure stage	The visual advances with real pressure; red is the ceiling warning state.

The current nine-program bank restores the canonical iCloud names without runtime clamping. VOCAL HOLD is value-identical to the preferred older preset; RED LINE WATCH is the strongest factory stage at 4.2 dB Maximize. The final safety curve uses raTanhPhi behavior rather than hard preset flattening.

Factory programs

NO BRICKS [Stereo]; VOCAL HOLD [Stereo]; BRICK TAP [Stereo]; WALL PRESSURE [Stereo]; SHOULDER CHECK [Stereo]; NO SCISSORS [Stereo]; BREAKTHROUGH [Stereo]; RED LINE WATCH [Stereo]; PHI GUARD [Stereo]

LOOKOUT

Dual-envelope transient/sustain shaper. Attack and Sustain directly move the level of the transient and sustain portions of the signal, up to +/-18 dB each, ahead of the DIH tone-color stage. Its Eye of Heru state also gives live advisory guidance on transient/sustain balance while it shapes.

Declared product: LookOut | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Insert Safe | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
power	Power	Factory
attack	Attack	Factory
sustain	Sustain	Factory
gainmatch	Gain Match	Factory

Artwork authority

assets/lookout_sentinel_bg.png

Meter and visualizer semantics

Display	What it measures	How to read it
Transient	Detected front-edge energy	Use to judge whether Attack is emphasizing or restraining hits.
Sustain	Detected body and tail energy	Use to judge Sustain changes to ring, room, and body.
Alert	Potential transient/sustain imbalance	Higher values suggest the balance needs attention.
Miss	Underrepresented event energy	Indicates material that may be disappearing.
Stress	Over-concentrated envelope pressure	Reduce shaping intensity when stress remains high.
Composure	Stable transient/sustain balance	Higher is calmer and more controlled.
Eye of Heru	Combined advisory state	It reports balance; it does not move controls automatically.

Factory programs

NEUTRAL SENTINEL [Stereo]; VOCAL FRONT GUARD; DRUM SNAP LOCK; KICK BODY ANCHOR; PERCUSSION EDGE; ROOM TAIL WATCH; MIX BUS COMPOSURE; FINAL PEAK SENTINEL